INF10025 Data Management and Analytics

Task 4 – Credit Submission

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A. Task Credit4a

According to the description, the task requires students to list the following columns named ***StudentID***, ***MovieNo***, ***movie title***, ***colour name***, and ***short rating description of every movie***. In addition, the list must be sorted in descending order of movie title. First of all, I need to check that all tables have been successfully created and verified in the database by using the command 'show tables;'. Then, I write the SQL statement and execute the query for task Credit4a.

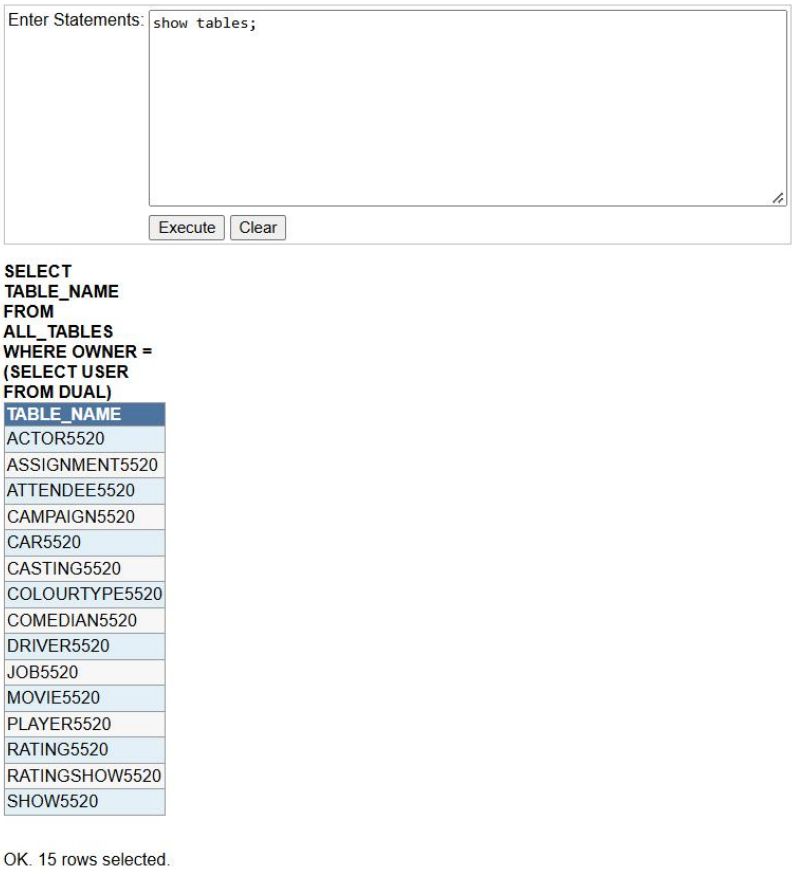


Figure 1: Verification of All Created Database Tables in iSQL JunioR.

Enter Statements:

```
SELECT '105565520' AS StuID, M.MOVIENO, M.TITLE, C.COLOURNAME,
R.SHORTDESC
FROM MOVIE5520 M
INNER JOIN COLOURTYPE5520 C ON M.COLOURCODE = C.COLOURCODE
INNER JOIN RATING5520 R ON M.RATINGCODE = R.RATINGCODE
ORDER BY TITLE DESC;
```

Execute Clear

Figure 2: SQL Query Statement for Task Credit4a.

SELECT '105565520' AS StuID, M.MOVIENO, M.TITLE, C.COLOURNAME, R.SHORTDESC FROM MOVIE5520 M INNER JOIN COLOURTYPE5520 C ON M.COLOURCODE = C.COLOURCODE INNER JOIN RATING5520 R ON M.RATINGCODE = R.RATINGCODE ORDER BY TITLE DESC

STUID	MOVIENO	TITLE	COLOURNAME	SHORTDESC
105565520	300000	the oracle of swinburne	Black & White Movie	GENERAL
105565520	329833	Zoolander 2	Colour Movie	MATURE (15+)
105565520	9398	Zoolander	Colour Movie	MATURE (15+)
105565520	338967	Zombieland: Double Tap	Colour Movie	MATURE ACCOMPANIED (15+)
105565520	19908	Zombieland	Colour Movie	MATURE (15+)
105565520	1819	You, Me and Dupree	Colour Movie	MATURE (15+)
105565520	127585	X-Men: Days of Future Past	Colour Movie	MATURE (15+)
105565520	246655	X-Men: Apocalypse	Colour Movie	MATURE (15+)
105565520	36657	X-Men	Colour Movie	MATURE (15+)
105565520	72190	World War Z	Colour Movie	MATURE (15+)
105565520	297762	Wonder Woman	Colour Movie	PARENTAL GUIDANCE
105565520	44147	Wild Target	Colour Movie	MATURE (15+)
105565520	11199	Wild Hogs	Colour Movie	MATURE (15+)
105565520	228970	Wild	Colour Movie	MATURE (15+)
105565520	356305	Why Him?	Colour Movie	MATURE (15+)

Figure 3: Output Result for Task Credit4a SQL Query.

B. Task Credit4b

In the next sub-task, students are required to write an SQL statement to list the ***stuid***, ***actor fullname, movie title***, and ***release year*** for each casting row in the Casting table. Notably, the list must be in descending order of actor fullname.

Enter Statements:

```
SELECT '105565520' AS StuID, A.FULLNAME, M.TITLE, M.RELYEAR
FROM MOVIE5520 M
INNER JOIN CASTING5520 C ON M.MOVIENO = C.MOVIENO
INNER JOIN ACTOR5520 A ON A.ACTORNO = C.ACTORNO
ORDER BY A.FULLNAME DESC;
```

Execute Clear

Figure 4: SQL Query Statement for Task Credit4b.

SELECT '105565520' AS StuID, A.FULLNAME, M.TITLE, M.RELYEAR FROM MOVIE5520 M INNER JOIN CASTING5520 C ON M.MOVIENO = C.MOVIENO INNER JOIN ACTOR5520 A ON A.ACTORNO = C.ACTORNO ORDER BY A.FULLNAME DESC

STUID	FULLNAME	TITLE	RELYEAR
105565520	Zoey Deutch	Vampire Academy	2014
105565520	Zoey Deutch	Zombieland: Double Tap	2019
105565520	Zoey Deutch	Before I Fall	2017
105565520	Zoey Deutch	Set It Up	2018
105565520	Zoey Deutch	The Disaster Artist	2017
105565520	Zoey Deutch	Why Him?	2016
105565520	Zoe Saldana	Star Trek Into Darkness	2013
105565520	Zoe Saldana	Avatar	2009
105565520	Zoe Saldana	Guardians of the Galaxy	2014
105565520	Zoe Saldana	Guardians of the Galaxy Vol. 2	2017
105565520	Zoe Saldana	Avengers: Endgame	2019
105565520	Zoe Saldana	The Words	2012
105565520	Zoe Saldana	Avengers: Infinity War	2018
105565520	Zoe Saldana	Guardians of the Galaxy Vol. 2	2017
105565520	Zoe Saldana	Pirates of the Caribbean: The Curse of the Black Pearl	2003

Figure 5: Output Result for Task Credit4b SQL Query.

C. Task Credit4c

In the third task, students have to write an SQL statement to l***ist the stuid, rating code, and the count of movies*** that belong to that rating code based on the movie table. In addition, ***the list should be in descending order of count.***

Enter Statements:

```
SELECT '105565520' AS StuID, RATINGCODE, COUNT(*) AS "Num of Movies"
FROM MOVIE5520
GROUP BY RATINGCODE
ORDER BY "Num of Movies" DESC;
```

Execute Clear

Figure 6: SQL Query Statement for Task Credit4c.

SELECT '105565520' AS StuID, RATINGCODE, COUNT(*) AS "Num of Movies" FROM MOVIE5520 GROUP BY RATINGCODE ORDER BY "Num of Movies" DESC

STUID	RATINGCODE	Num of Movies
105565520	M	181
105565520	PG	120
105565520	MA	78
105565520	G	16

OK. 4 rows selected.

Figure 7: Output Result for Task Credit4c SQL Query.

D. Task Credit4d

Similar to task Credit4c, the description adds a new requirement to only include ***results where the count is greater than 21***.

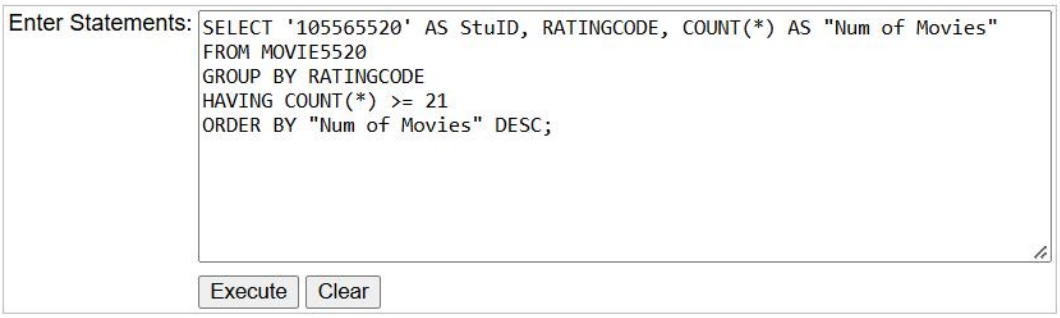
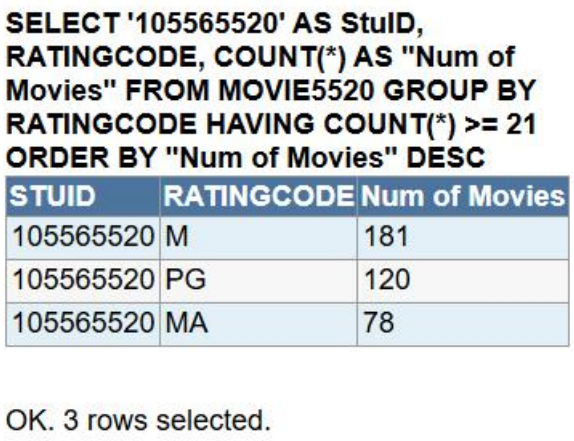


Figure 8: SQL Query Statement for Task Credit4d.



SELECT '105565520' AS StuID, RATINGCODE, COUNT(*) AS "Num of Movies" FROM MOVIE5520 GROUP BY RATINGCODE HAVING COUNT(*) >= 21 ORDER BY "Num of Movies" DESC

STUID	RATINGCODE	Num of Movies
105565520	M	181
105565520	PG	120
105565520	MA	78

OK. 3 rows selected.

Figure 9: Output Result for Task Credit4d SQL Query.

E. Task Credit4e

Next, for task Credit4e, students are required to write an SQL statement that ***counts the movies by rating code,*** with an additional f***ilter restricting the records to movies released between 1996 and 2010 (inclusive)***.

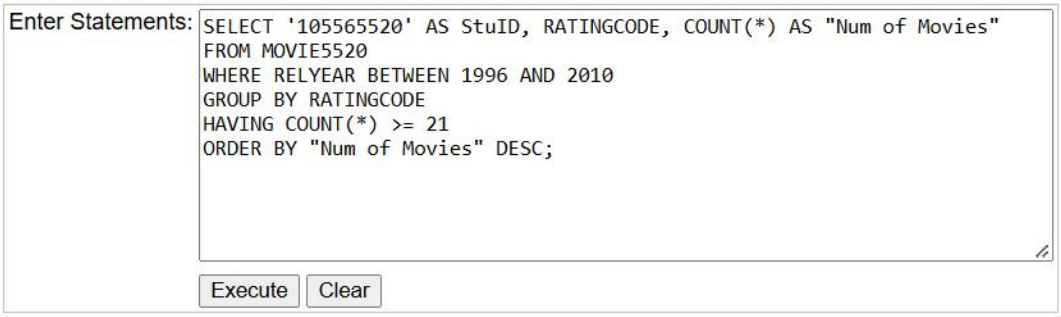


Figure 10: SQL Query Statement for Task Credit4e.

SELECT '105565520' AS StuID, RATINGCODE, COUNT(*) AS "Num of Movies" FROM MOVIE5520 WHERE RELYEAR BETWEEN 1996 AND 2010 GROUP BY RATINGCODE HAVING COUNT(*) >= 21 ORDER BY "Num of Movies" DESC

STUID	RATINGCODE	Num of Movies
105565520	M	80
105565520	PG	45
105565520	MA	30

OK. 3 rows selected.

Figure 11: Output Result for Task Credit4e SQL Query.

F. Task Credit4f

In task Credit4f, students are required to write an SQL statement to ***list the stuid, actor fullname, gender, and the total number of movies they have appeared in, with the column heading renamed to 'Total Movies'.*** In addition, the ***resulting list must be sorted in descending order of actor fullname.***

Enter Statements:

```
SELECT '105565520' AS StuID, A.FULLNAME, A.GENDER, COUNT(*) AS "Total
Movies"
FROM ACTOR5520 A
INNER JOIN CASTING5520 C ON A.ACTORNO = C.ACTORNO
INNER JOIN MOVIE5520 M ON C.MOVIENO = M.MOVIENO
GROUP BY A.FULLNAME, A.GENDER, A.ACTORNO
ORDER BY A.FULLNAME DESC;
```

Execute Clear

Figure 12: SQL Query Statement for Task Credit4f.

```
SELECT '105565520' AS StuID, A.FULLNAME, A.GENDER,
COUNT(*) AS "Total Movies" FROM ACTOR5520 A INNER
JOIN CASTING5520 C ON A.ACTORNO = C.ACTORNO
INNER JOIN MOVIE5520 M ON C.MOVIENO = M.MOVIENO
GROUP BY A.FULLNAME, A.GENDER, A.ACTORNO
ORDER BY A.FULLNAME DESC
```

STUID	FULLNAME	GENDER	Total Movies
105565520	Zoey Deutch	F	6
105565520	Zoe Saldana	F	9
105565520	Zoe Kazan	F	3
105565520	Zeljko Ivanek	M	5
105565520	Zazie Beetz	F	1
105565520	Zach Woods	M	3
105565520	Zach Galifianakis	M	3
105565520	Zac Efron	M	4
105565520	Woody Harrelson	M	8
105565520	Winona Ryder	F	4
105565520	Willow Shields	F	4
105565520	Willie Garson	M	3
105565520	William Sadler	M	3
105565520	William Melling	M	5
105565520	William Hurt	M	4

Figure 13: Output Result for Task Credit4f SQL Query.

G. Task Credit4g

In task Credit4g, students are required to write an SQL statement to ***list the stuid, actor fullname, rating code, and the number of movies they appeared in for that code.*** Furthermore, the resulting l***ist must be sorted in ascending order by actor fullname and rating code.***

Enter Statements:

```
SELECT '105565520' AS StuID, A.FULLNAME, M.RATINGCODE, COUNT(*) AS "Num
of Movies"
FROM ACTOR5520 A
INNER JOIN CASTING5520 C ON C.ACTORNO = A.ACTORNO
INNER JOIN MOVIE5520 M ON M.MOVIENO = C.MOVIENO
GROUP BY A.FULLNAME, M.RATINGCODE
ORDER BY A.FULLNAME, M.RATINGCODE ASC;
```

Execute Clear

Figure 14: SQL Query Statement for Task Credit4g.

```
SELECT '105565520' AS StuID, A.FULLNAME, M.RATINGCODE,
COUNT(*) AS "Num of Movies" FROM ACTOR5520 A INNER JOIN
CASTING5520 C ON C.ACTORNO = A.ACTORNO INNER JOIN
MOVIE5520 M ON M.MOVIENO = C.MOVIENO GROUP BY
A.FULLNAME, M.RATINGCODE ORDER BY A.FULLNAME,
M.RATINGCODE ASC
```

STUID	FULLNAME	RATINGCODE	Num of Movies
105565520	Aaron Himelstein	M	3
105565520	Aaron Himelstein	MA	1
105565520	Abigail Breslin	M	3
105565520	Abigail Breslin	PG	1
105565520	Adam Devine	M	5
105565520	Adam Devine	PG	1
105565520	Adam Garcia	M	1
105565520	Adam Garcia	PG	1
105565520	Adam Sandler	G	1
105565520	Adam Sandler	M	3
105565520	Adrian Rawlins	M	2
105565520	Adrian Rawlins	PG	5
105565520	Afshan Azad	M	2
105565520	Afshan Azad	PG	3
105565520	Al Pacino	MA	2

Figure 15: Output Result for Task Credit4g SQL Query.

H. Task Credit4h

In task Credit4h, students are required to download the SQL script named ***task04_customers.txt*** and modify it by replacing **all occurrences of customerXXXX with customer5520 (utilizing the last 4 digits of my student ID)**. Subsequently, an SQL statement must be constructed to calculate the sum of all sales this year grouped by state and gender, including only customers with a rating code of 1, 2, or 3. Furthermore, the resulting list must be sorted in ascending order by state and gender sequence.

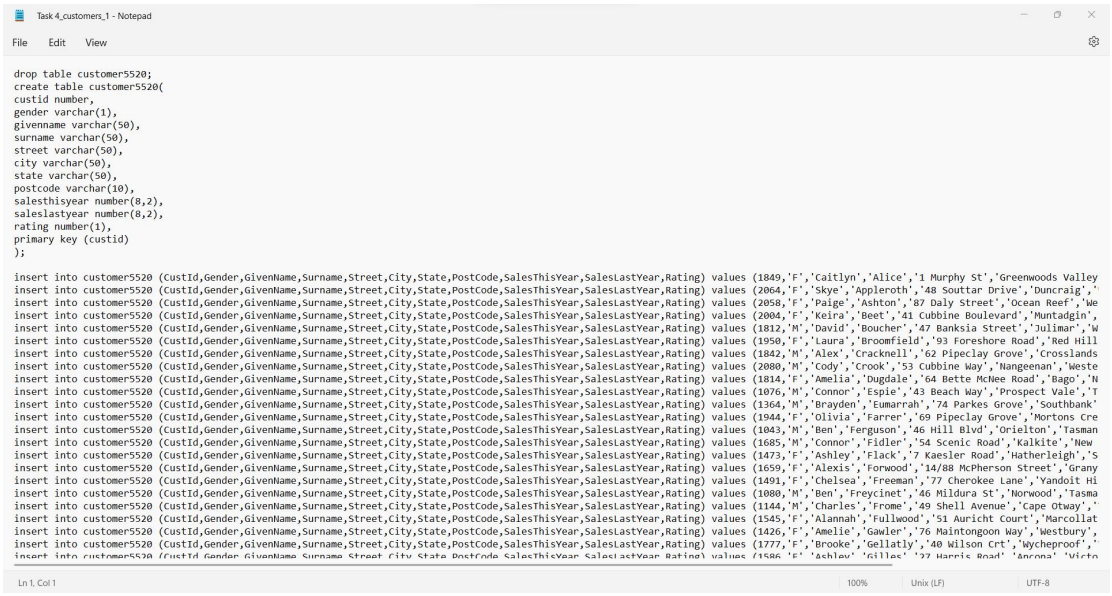


Figure 16: Table Creation and Data Insertion Script for customer5520.

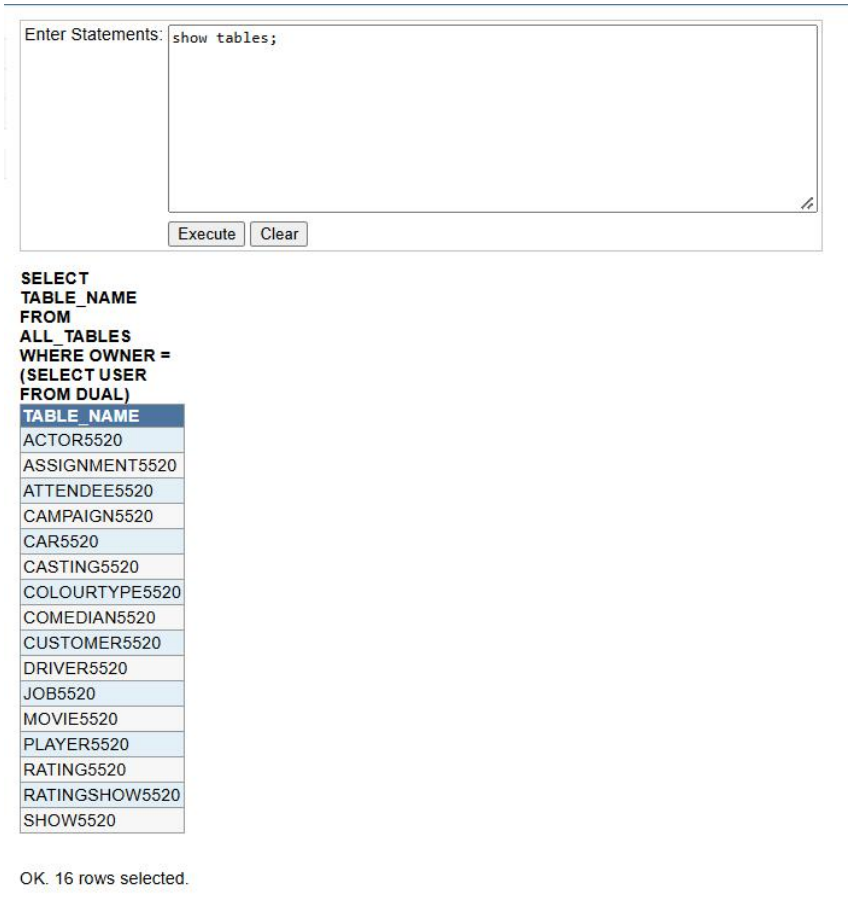


Figure 17: List of Database Tables Verified in iSQL JunioR.

Enter Statements:
```
SELECT * FROM CUSTOMER5520
```
Execute Clear

SELECT * FROM CUSTOMER5520

CUSTID	GENDER	GIVENNAME	SURNAME	STREET	CITY	STATE	POSTCODE	SALESTHISYEAR	SALESLASTYEAR	RATING
1849	F	Caitlyn	Alice	1 Murphy St	Greenwoods Valley	Western Australia	6405	1379	1793	2
2064	F	Skye	Appleroth	48 Souttar Drive	Duncraig	Western Australia	6023	1151	1221	2
2058	F	Paige	Ashton	87 Daly Street	Ocean Reef	Western Australia	6027	3863	1932	2
2004	F	Keira	Beet	41 Cubbine Boulevard	Muntadgin	Western Australia	6420	687	83	1
1812	M	David	Boucher	47 Banksia Street	Julimar	Western Australia	6567	1744	1082	2
1950	F	Laura	Broomfield	93 Foreshore Road	Red Hill	Western Australia	6056	0	0	1
1842	M	Alex	Cracknell	62 Pipeclay Grove	Crosslands	New South Wales	2446	288	121	1
2080	M	Cody	Crook	53 Cubbine Way	Nangeenan	Western Australia	6414	32	28	2
1814	F	Amelia	Dugdale	64 Bette McNee Road	Bago	New South Wales	2730	0	0	2
1076	M	Connor	Espie	43 Beach Way	Prospect Vale	Tasmania	7250	118	202	1
1364	M	Brayden	Eumarrah	74 Parkes Grove	Southbank	Victoria	3006	53	101	2
1944	F	Olivia	Farrer	69 Pipeclay Grove	Mortons Creek	New South Wales	2446	598	664	2
1043	M	Ben	Ferguson	46 Hill Blvd	Orielton	Tasmania	7172	0	0	3
1685	M	Connor	Fidler	54 Scenic Road	Kalkite	New South Wales	2627	3015	2081	1
1473	F	Ashley	Flack	7 Kaesler Road	Hatherleigh	South Australia	5280	725	1385	1
1659	F	Alexis	Forwood	14/88 McPherson Street	Granya	Victoria	3701	603	157	2

Figure 18: Data Verification for Table CUSTOMER5520.

Enter Statements:
```
SELECT '105565520' AS StuID, STATE, GENDER, SUM(SALESTHISYEAR) AS "Total
Sales"
FROM CUSTOMER5520
WHERE RATING IN (1, 2, 3)
GROUP BY STATE, GENDER
ORDER BY STATE, GENDER ASC;
```
Execute Clear

Figure 19: SQL Query Statement for Task Credit 4h.

SELECT '105565520' AS StuID, STATE, GENDER, SUM(SALESTHISYEAR) AS "Total Sales" FROM CUSTOMER5520 WHERE RATING IN (1, 2, 3) GROUP BY STATE, GENDER ORDER BY STATE, GENDER ASC

STUID	STATE	GENDER	Total Sales
105565520	New South Wales	F	899
105565520	New South Wales	M	3386
105565520	South Australia	F	2430
105565520	Tasmania	F	1312
105565520	Tasmania	M	229
105565520	Victoria	F	6695
105565520	Victoria	M	7001
105565520	Western Australia	F	11247
105565520	Western Australia	M	3169

OK. 9 rows selected.

Figure 20: Output Result for Task Credit4h SQL Query.

It is clear from the Figure 20 that the query results show the total sales this year (SALESTHISYEAR) grouped by state and gender for customers with a rating code of 1, 2, or 3. Overall, the data indicates variations in purchasing performance across different regions, with certain states exhibiting significantly higher total sales than others. Additionally, the breakdown by gender reveals distinct spending distributions within each state, providing valuable insights into customer purchasing behaviors based on their demographics and ratings.

Conclusion

In conclusion, this assignment successfully applied core Oracle SQL concepts within the iSQL JunioR environment, including:

- Table management and schema constraints standardized with the 5520 suffix.
- Advanced querying utilizing INNER JOIN operations, aggregate functions (COUNT, SUM), and filtering conditions (WHERE, IN).
- Data grouping and sorting (GROUP BY, ORDER BY) to extract accurate analytical reports regarding actors, movies, and customer sales by region.

Overall, the project effectively reinforced practical relational database management and querying skills.